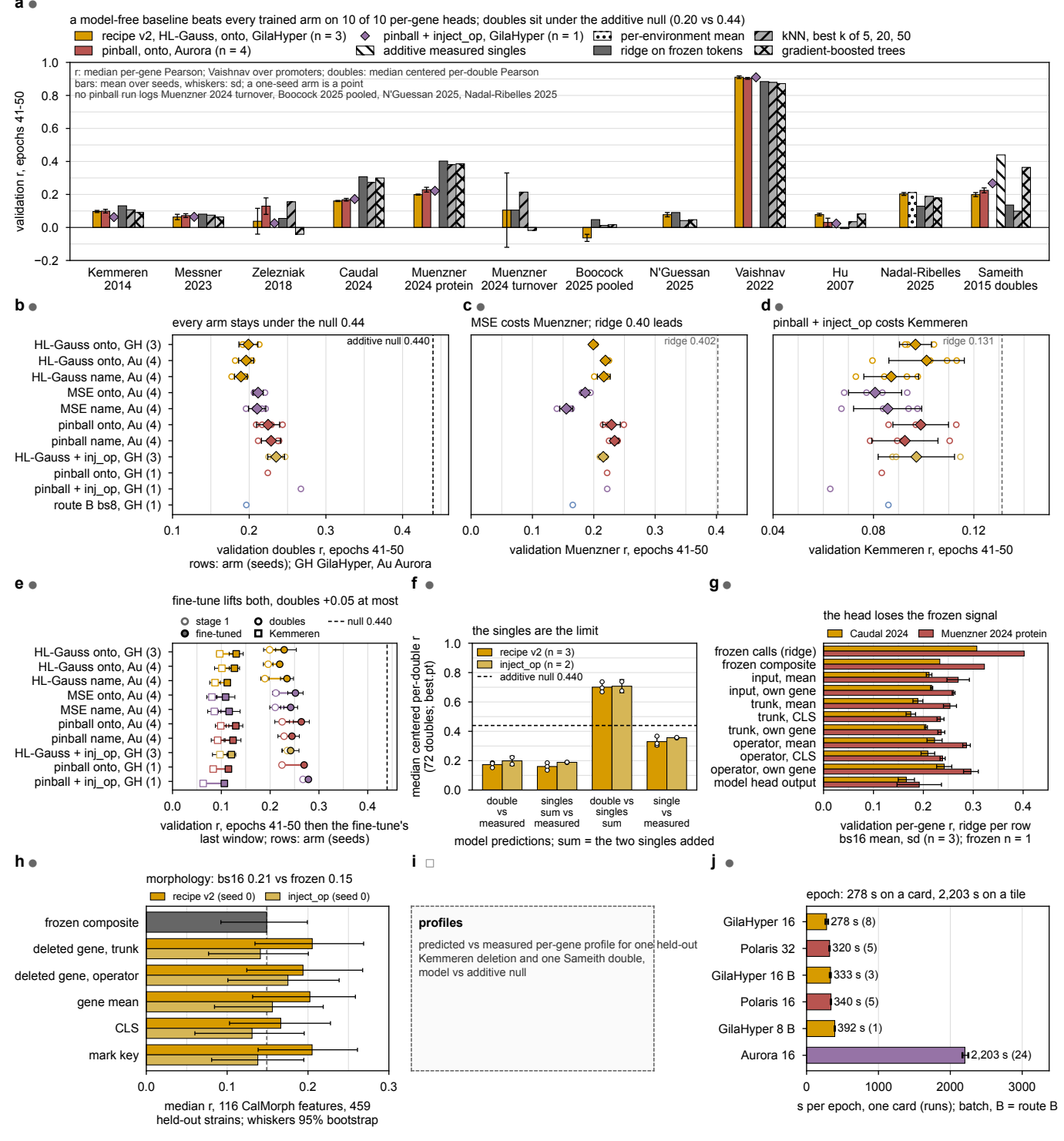

a
a model-free baseline beats every trained arm on 10 of 10 per-gene heads; doubles sit under the additive null (0.20 vs 0.44)
recipe v2, HL-Gauss, onto, GilaHyper (n = 3)
pinball, onto, Aurora (n = 4)
pinball + inject_op, GilaHyper (n = 1)
additive measured singles
per-environment mean
ridge on frozen tokens
kNN, best k of 5, 20, 50
gradient-boosted trees
r: median per-gene Pearson; Vaishnav over promoters; doubles: median centered per-double Pearson
bars: mean over seeds, whiskers: sd; a one-seed arm is a point
no pinball run logs Muenzner 2024 turnover, Boocock 2025 pooled, N'Guessan 2025, Nadal-Ribelles 2025
validation r, epochs 41-50
Kemmeren 2014
Messner 2023
Zelezniak 2018
Caudal 2024
Muenzner 2024 protein
Muenzner 2024 turnover
Boocock 2025 pooled
N'Guessan 2025
Vaishnav 2022
Hu 2007
Nadal-Ribelles 2025
Sameith 2015 doubles
b
every arm stays under the null 0.44
additive null 0.440
HL-Gauss onto, GH (3)
HL-Gauss onto, Au (4)
HL-Gauss name, Au (4)
MSE onto, Au (4)
MSE name, Au (4)
pinball onto, Au (4)
pinball name, Au (4)
HL-Gauss + inj_op, GH (3)
pinball onto, GH (1)
pinball + inj_op, GH (1)
route B bs8, GH (1)
validation doubles r, epochs 41-50
rows: arm (seeds); GH GilaHyper, Au Aurora
c
MSE costs Muenzner; ridge 0.40 leads
ridge 0.402
validation Muenzner r, epochs 41-50
d
pinball + inj_op costs Kemmeren
ridge 0.131
validation Kemmeren r, epochs 41-50
e
fine-tune lifts both, doubles +0.05 at most
stage 1
fine-tuned
doubles
Kemmeren
null 0.440
validation r, epochs 41-50 then the fine-tune's last window; rows: arm (seeds)
f
the singles are the limit
recipe v2 (n = 3)
inject_op (n = 2)
additive null 0.440
median centered per-double r (72 doubles; best.pt)
double vs measured
singles sum vs measured
double vs singles sum
single vs measured
model predictions; sum = the two singles added
g
the head loses the frozen signal
Caudal 2024
Muenzner 2024 protein
frozen calls (ridge)
frozen composite
input, mean
input, own gene
trunk, mean
trunk, CLS
trunk, own gene
operator, mean
operator, CLS
operator, own gene
model head output
validation per-gene r, ridge per row
bs16 mean, sd (n = 3); frozen n = 1
h
morphology: bs16 0.21 vs frozen 0.15
recipe v2 (seed 0)
inject_op (seed 0)
frozen composite
deleted gene, trunk
deleted gene, operator
gene mean
CLS
mark key
median r, 116 CalMorph features, 459 held-out strains; whiskers 95% bootstrap
i
profiles
predicted vs measured per-gene profile for one held-out Kemmeren deletion and one Sameith double, model vs additive null
j
epoch: 278 s on a card, 2,203 s on a tile
GilaHyper 16
Polaris 32
GilaHyper 16 B
Polaris 16
GilaHyper 8 B
Aurora 16
278 s (8)
320 s (5)
333 s (3)
340 s (5)
392 s (1)
2,203 s (24)
s per epoch, one card (runs); batch, B = route B